PhysAgent: Physics-Informed SDevice Code Generation via Ontology-Guided LLM

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***Abstract*—As semiconductor scaling reaches its limits, TCAD simulation becomes essential for DTCO. However, the core challenge lies in configuring SDevice physics models and parameters, not in structure generation. We propose PhysAgent, which builds a Sentaurus TCAD guide-based ontology and integrates literature-driven domain knowledge to generate physically grounded SDevice files without relying on synthetic training data.**

***Keywords*—TCAD, ontology, physics-informed modeling, parameter file generation, domain knowledge, semiconductor materials**

I. Introduction

Recent studies have applied LLMs to automate TCAD code generation, primarily targeting SDE structure files. These approaches rely on supervised fine-tuning (SFT) with synthetic datasets constructed by varying numerical parameters in a small number of baseline scripts [3]–[6]. However, this limits generalization to unseen device architectures or materials. More critically, none adequately addresses physics model selection or parameter file generation for SDevice—the core of physically meaningful TCAD simulation. Even AgenticTCAD [6], which generates SDevice code, uses only 2–3 fixed physics configurations and omits parameter files entirely.

We propose PhysAgent, which constructs a Sentaurus TCAD guide-based ontology encoding physical models, equations, and their interdependencies, complemented by literature-driven material-specific knowledge. By providing this structured context to general-purpose LLMs, PhysAgent generates physically grounded SDevice files with explicit reasoning about model selection and parameter settings—without domain-specific fine-tuning.

II. Related Work

TCAD simulation consists of SDE for device geometry and SDevice for physics models and solver settings. While LLMs have been applied to SDE code generation through fine-tuning or in-context learning, SDevice configuration—selecting and parameterizing models such as mobility, recombination, and bandgap narrowing—remains unaddressed due to its high expertise requirements.

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III. Proposed Framework

PhysAgent combines a TCAD physics ontology and literature-driven domain knowledge to guide LLM-based SDevice file generation. The constructed ontology is publicly available as an interactive web platform [7].

A. TCAD Physics Ontology

We construct a directed graph from the Sentaurus TCAD guide with three node types (*equation*, *parameter*, *constant*) and three edge types (*input*, *specialCase*, *pairwise*), capturing inter-model dependencies such as bandgap narrowing affecting intrinsic carrier density and recombination rates (Fig. 1).

B. Literature-Driven Domain Knowledge

The ontology captures model structure but not material-specific expertise. PhysAgent supplements it with literature-driven knowledge—validated model combinations and calibrated parameters for specific materials (e.g., SiC, GaN).

C. Ontology-Guided Generation

Given a target device and material, PhysAgent extracts a relevant ontology subgraph via MCP (Model Context Protocol) tool calls, combines it with literature knowledge, and provides this as structured context to the LLM, producing physically justified SDevice configurations (Fig. 2).

D. Ontology Expansion and Validation

Unlike SFT-based approaches requiring retraining for new models or materials, PhysAgent uses a directional synchronization workflow (Fig. 3) with three artifact layers—Spec, Plan, and Impl—where changes propagate automatically from the source of truth while reverse modifications require explicit approval, ensuring incremental expansion without retraining.

IV. Result

We evaluate PhysAgent through three experiments.

Experiment 1: Material-Specific Generation. We generate SDevice parameter files for two library materials (SiC, GaN) and one non-library material (IGZO). For SiC and GaN, we compare the generated files against Sentaurus TCAD's built-in library models to assess physical accuracy (Fig. 4). For IGZO, which has no library model, we demonstrate PhysAgent's ability to construct parameter files from literature alone, validating extensibility beyond pre-existing templates.

Experiment 2: Ablation Study. We compare two configurations: (1) a baseline where the LLM receives only the Sentaurus guide text without ontology or MCP tools, and (2) PhysAgent with ontology-guided subgraph extraction via MCP and literature-driven knowledge. We measure model completeness (fraction of required physics models included), parameter accuracy (deviation from literature values), and inter-model consistency (Fig. 5).

Experiment 3: Automated Validation. We verify generated files through three automated checks: ontology compliance (no missing required parameters), syntax validation (executable by Sentaurus), and physics validation (parameter ranges and model compatibility within ontology constraints).

V. Conclusion

We presented PhysAgent, which generates physically grounded SDevice files by combining a TCAD guide-based ontology with literature-driven domain knowledge, enabling general-purpose LLMs to support expert-level TCAD modeling without synthetic training data.

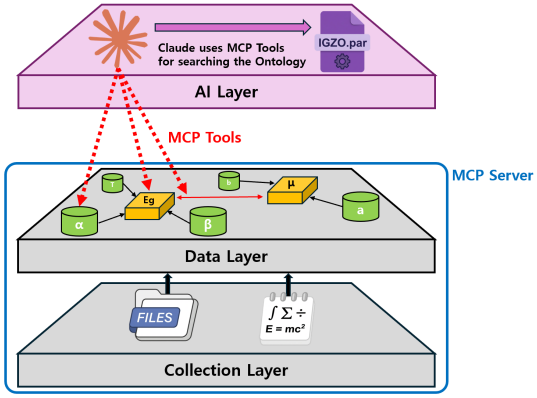

Fig. 1. Ontology system architecture.

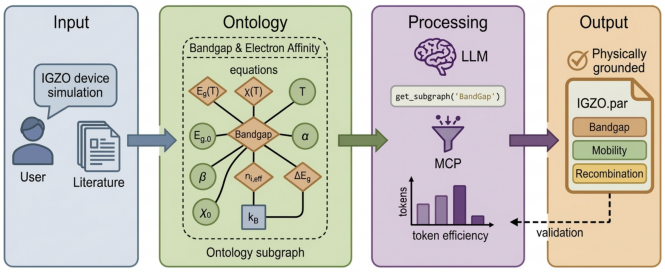

Fig. 2. PhysAgent framework: LLM agents access the ontology graph via MCP tool calls to generate physically grounded parameter files.

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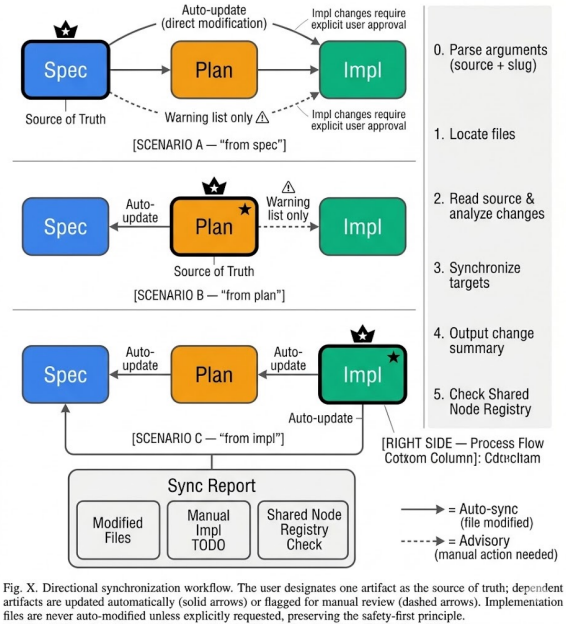

Fig. X. Directional synchronization workflow. The user designates one artifact as the source of truth; dependent artifacts are updated automatically (solid arrows) or flagged for manual review (dashed arrows). Implementation files are never auto-modified unless explicitly requested, preserving the safety-first principle.

Fig. 3. Expansion and validation workflow.

[Placeholder: Experiment 1 — SiC, GaN, IGZO parameter files vs library models]

Fig. 4. Experiment 1: Material-specific parameter file generation.

[Placeholder: Experiment 2 — baseline vs PhysAgent ablation results]

Fig. 5. Experiment 2: Ablation study comparing baseline LLM and PhysAgent.

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